

Faculty of Computing



[Computer Communications & Network]

Lab No 5 Tasks

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Task 1: Use the IP address chart and your knowledge of IP address classes to answer the following questions:

1. **What is the decimal and binary range of the first octet of all possible Class B IP addresses?**

□ Decimal: From: 128 To: 191

□ Binary: From: 10000000 To: 10111111

2. **Which octet(s) represent the network portion of a Class C IP address? 3 octets at starting.**

3. **Which octet(s) represent the host portion of a Class A IP address? Last 3 octets**

4. **What is the maximum number of useable hosts with a Class C network address? $2^8 - 2 = 254$**

5. **How many Class B networks are there? 16384**

6. **How many hosts can each Class B network have? 65534**

7. **How many octets are there in an IP address? 4 How many bits per octet? 8**

Task 2: Determine the host and network portions of the IP address

With the following IP host addresses, indicate the following:

□ Class of each address

□ Network address or ID

□ Host portion

□ Default subnet mask

The host portion will be all zeros for the network ID. Enter just the octets that make up the host. The host portion will be all ones for a broadcast. The network portion of the address will be all ones for the subnet mask. Fill in the following table:

Host IP Address	Address Class	Network Address	Host Address	Default Subnet Mask
216.14.55.137	C	216.14.55.0	137	255.255.255.0
123.1.1.15	A	123.0.0.0	1.1.15	255.0.0.0
150.127.221.244	B	150.127.0.0	221.244	255.255.0.0
194.125.35.199	C	194.125.35.0	199	255.255.255.0
175.12.239.244	B	175.12.0.0	239.244	255.255.0.0

Task 3: Given an IP address of 142.226.0.15, answer the following questions:

What is the binary equivalent of the second octet? 111000010

What is the class of the address? Class B

What is the network address of this IP address? 142.226.0.0

Is this a valid IP host address
(Y/N)? Why or why not?

Yes it is valid because it belongs to class B

It is not network address.

It is not a broadcast address.

Task 4: Determine which IP host addresses are valid for commercial networks

IP Host Address	Valid Address? (Yes/No)	Why or Why Not
150.100.255.255	NO	Because it is not belonging to any class
175.100.255.18	YES	It belongs to class B
195.234.253.0	NO	Last oct 0 so it is not belong to c as c has 3 network and 1 host
100.0.0.23	YES	Class A hots because(1-126)
188.258.221.176	NO	258 is less than 255 so invalid
127.34.25.189	NO	Reserved loopback
224.156.217.73	NO	It belongs to class d which is multicast